

ON PRIMES p FOR WHICH $N - p$ IS AN ALMOST PRIME SUM OF TWO SQUARES

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ABSTRACT. Let N be a sufficiently large even integer. We show that the number of primes $p \leq N$ such that $N - p$ is a sum of two squares having at most nine prime factors, counted with multiplicity, is $\gg \mathfrak{S}(N)N/(\log N)^2$. The proof follows the vector sieve framework of Nath and Xie, combining two semi-linear sieves at the Bombieri–Vinogradov level of distribution with Richert’s logarithmic weights. Integers with two large prime factors $\equiv 3 \pmod{4}$, which escape the semi-linear sieve, are removed by a switching argument. The final numerical inequality is verified with outward-rounded interval arithmetic.

1. INTRODUCTION

Chen’s theorem [4] states that every sufficiently large even integer N is of the form $N = p + P_2$, where P_2 has at most two prime factors. Linnik [9] showed that every sufficiently large integer is of the form $p + x^2 + y^2$, and Iwaniec [6] and Motohashi [10] studied primes p with $p - 1$ a sum of two squares by sieve methods; the semi-linear (half-dimensional) sieve of Iwaniec [7] is the natural tool for the condition “ n is a sum of two squares”. More recently Nath and Xie [12] combined a semi-linear and a linear sieve in a vector sieve [3] to show that there are $\gg x/(\log x)^{5/2}$ primes $p \leq x$ such that $p - 1$ is a sum of two squares and $\Omega(p + 2) \leq 11$.

In this note we consider the additive problem in which both conditions are imposed on the *same* shifted number $N - p$.

Theorem 1.1. *There is an absolute constant N_0 such that for every even $N \geq N_0$,*

$$\#\{p \leq N : N - p = x^2 + y^2 \text{ for some } x, y \in \mathbb{Z}, \Omega(N - p) \leq 9\} \geq \frac{1}{10} \mathfrak{S}(N) \frac{N}{(\log N)^2},$$

where $\mathfrak{S}(N) = 2 \prod_{\ell > 2} (1 - (\ell - 1)^{-2}) \prod_{\ell | N, \ell > 2} \frac{\ell - 1}{\ell - 2}$ is the binary Goldbach singular series. The constant N_0 is ineffective.

Method. Write $n = N - p$ and restrict to $p \equiv N - 1 \pmod{4}$, so that $n \equiv 1 \pmod{4}$. If n has no prime factor $q \equiv 3 \pmod{4}$ at all, then n is a sum of two squares. With the Bombieri–Vinogradov level $N^{1/2}$ one cannot sift the primes $q \equiv 3 \pmod{4}$ all the way up to $N^{1/2}$: the semi-linear lower bound function satisfies $f_{1/2}(s) = 0$ for $s \leq 1$. We therefore sift them only up to $y = N^{1/2-\delta}$. Since $n \equiv 1 \pmod{4}$, a survivor then has either no prime factor $\equiv 3 \pmod{4}$, or exactly two, both $\geq y$; the latter are counted from above by switching the roles of p and $q_1 q_2 m$ (Section 4). The number of prime factors is controlled by Richert’s logarithmic weights (Section 3). The resulting numerical inequality is verified rigorously in Section 5.

Remark 1.2. To the best of the author’s knowledge, the explicit statement of Theorem 1.1 has not appeared in the literature. The analytic architecture is an adaptation of the vector sieve method of Nath and Xie [12]. The ingredient specific to the present setting is the treatment of the $s = 1$ threshold of the semi-linear sieve by truncated sieving at $y = N^{1/2-\delta}$, followed by the switching step of Section 4.

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2. NOTATION AND TOOLS

Throughout, N is a large even integer, $\varepsilon > 0$ is small, p, q, ℓ denote primes, $\mathcal{P}_1 = \{q \equiv 1 \pmod{4}\}$, $\mathcal{P}_3 = \{q \equiv 3 \pmod{4}\}$, and for a set of primes $\mathcal{P}$ we write $P(\mathcal{P}; z) = \prod_{q \in \mathcal{P}, q < z} q$. Let

$$\mathcal{A} = \{n = N - p : p \leq N, p \equiv N - 1 \pmod{4}\}, \quad X = \frac{1}{2} \text{li}(N).$$

For odd squarefree d put $g(d) = \prod_{\ell|d} g(\ell)$ with $g(\ell) = 1/(\ell - 1)$ if $\ell \nmid N$ and $g(\ell) = 0$ if $\ell \mid N$, and define r_d by $\#\{n \in \mathcal{A} : d \mid n\} = Xg(d) + r_d$.

Lemma 2.1 (Bombieri–Vinogradov with divisor weights). *For every $A > 0$ there is $B = B(A)$ such that, with $D = N^{1/2}(\log N)^{-B}$,*

$$\sum_{d \leq D} \mu^2(d) \tau_3(d) |r_d| \ll_A \frac{N}{(\log N)^A}.$$

Proof. For odd d with $(d, N) = 1$, $\#\{n \in \mathcal{A} : d \mid n\} = \pi(N; 4d, a_d)$ for a residue a_d coprime to $4d$, and $\varphi(4d) = 2\varphi(d)$; for $(d, N) > 1$ the count is $O(1)$. Hence $|r_d| \leq E(N; 4d) + O(1)$ with $E(N; k) = \max_{(a, k)=1} |\pi(N; k, a) - \text{li}(N)/\varphi(k)|$. By Brun–Titchmarsh, $|r_d| \ll N/(\varphi(d) \log N)$ for $d \leq D$. By Cauchy–Schwarz,

$$\sum_{d \leq D} \tau_3(d) |r_d| \leq \left(\sum_{d \leq D} \tau_3(d)^2 |r_d| \right)^{1/2} \left(\sum_{d \leq D} |r_d| \right)^{1/2} \ll \left(N(\log N)^C \right)^{1/2} \left(\frac{N}{(\log N)^{2A+C}} \right)^{1/2},$$

using $\sum_{d \leq D} \tau_3(d)^2 / \varphi(d) \ll (\log N)^C$ and the Bombieri–Vinogradov theorem [1] for the moduli $4d \leq 4D$. $\square$

Lemma 2.2 (The semi-linear sieve). *Let $F = F_{1/2}$, $f = f_{1/2}$ be the continuous solutions of*

$$(\sqrt{s} F(s))' = \frac{1}{2\sqrt{s}} f(s-1), \quad (\sqrt{s} f(s))' = \frac{1}{2\sqrt{s}} F(s-1) \quad (s > 1),$$

with $F(s) = A/\sqrt{s}$ for $0 < s \leq 2$, $f(s) = 0$ for $s \leq 1$, and $A = 2\sqrt{e^\gamma/\pi}$. Then $f(s) = A \text{arccosh}(\sqrt{s})/\sqrt{s}$ for $1 \leq s \leq 3$, F is decreasing, f is increasing, and $F(s), f(s) \rightarrow 1$. Let $\mathcal{P} \in \{\mathcal{P}_1, \mathcal{P}_3\}$. For $z \leq D$ and $s = \log D / \log z$ there are upper and lower sieve weights $\lambda_d^\pm$, $|\lambda_d^\pm| \leq 1$, supported on $d \mid P(\mathcal{P}; z)$, $d \leq D$, with $\sum_{d|(n, P(\mathcal{P}; z))} \lambda_d^- \leq 1_{(n, P(\mathcal{P}; z))=1} \leq \sum_{d|(n, P(\mathcal{P}; z))} \lambda_d^+$ and

$$\sum_d \lambda_d^\pm g(d) = V(\mathcal{P}; z) (F(s) \text{ resp. } f(s) + o(1)), \quad V(\mathcal{P}; z) = \prod_{q \in \mathcal{P}, q < z} (1 - g(q)).$$

Proof. This is Iwaniec’s β -sieve with $\kappa = 1/2$, $\beta = 1$ [7, 8]; see also [5, Ch. 11, 14]. The sets $\mathcal{P}_1, \mathcal{P}_3$ have sifting density $1/2$ by the prime number theorem in progressions modulo 4. The closed form on $[1, 3]$ follows by integrating the second equation. The asymptotic evaluation of the main terms follows the standard formulation of Iwaniec’s half-dimensional sieve; see [12, Prop. 3.10] and [5, Ch. 14]. $\square$

Figure 1 shows $F_{1/2}$ and $f_{1/2}$. Since $f_{1/2}(s) = 0$ for $s \leq 1$, sifting the primes $q \equiv 3 \pmod{4}$ up to $N^{1/2}$ with level $N^{1/2}$ gives no lower bound; this is why we sift them only up to $N^{1/2-\delta}$ in Section 3.

Lemma 2.3 (Vector sieve). *Let $a, b \in \{0, 1\}$ and $a^- \leq a \leq a^+$, $b^- \leq b \leq b^+$ with $a^+, b^+ \geq 0$. Then $ab \geq a^-b^+ + a^+b^- - a^+b^+$ and $ab \leq a^+b^+$.*

Proof. $(a^+ - a)(b^+ - b) \geq 0$ gives $ab \geq a^+b + ab^+ - a^+b^+$, and $a^+b \geq a^+b^-$, $ab^+ \geq a^-b^+$. $\square$

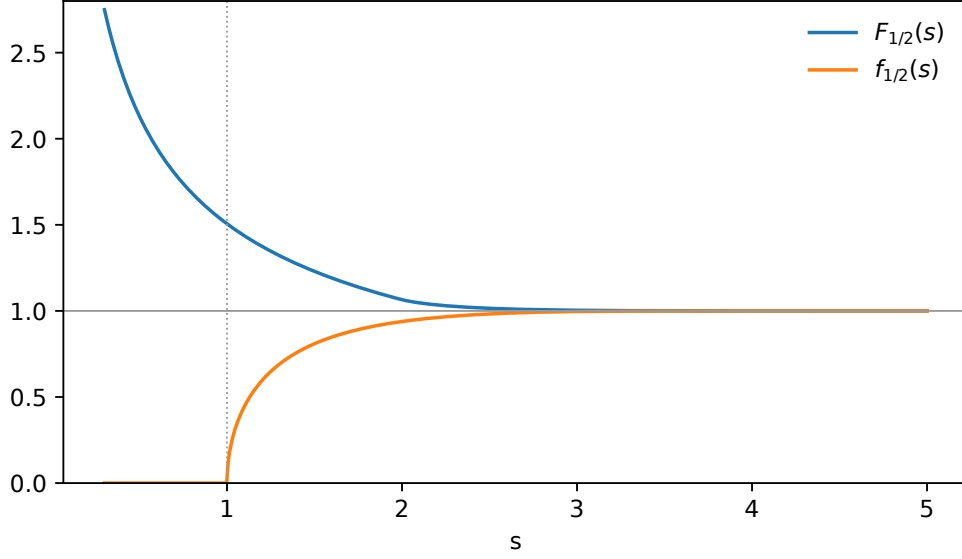


FIGURE 1. The semi-linear sieve functions $F_{1/2}(s)$ and $f_{1/2}(s)$ (dimension $\kappa = 1/2$, sieving limit $\beta = 1$). The lower bound function vanishes for $s \leq 1$, so at the Bombieri–Vinogradov level $\theta = 1/2$ the primes $q \equiv 3 \pmod{4}$ cannot be sifted all the way up to $N^{1/2}$.

Lemma 2.4 (Bombieri–Vinogradov for convolutions). *Let α_k, β_l be complex coefficients bounded by divisor functions, supported on $K < k \leq 2K$ and $l \leq N/K$ with $N^\varepsilon \leq K \leq N^{1-\varepsilon}$, and suppose β satisfies the Siegel–Walfisz condition. Then for every $A > 0$ there is B with*

$$\sum_{d \leq N^{1/2}(\log N)^{-B}} \max_{(a,d)=1} \left| \sum_{\substack{kl \leq N \\ kl \equiv a \pmod{d}}} \alpha_k \beta_l - \frac{1}{\varphi(d)} \sum_{\substack{kl \leq N \\ (kl,d)=1}} \alpha_k \beta_l \right| \ll \frac{N}{(\log N)^A}.$$

Proof. The result follows from the bilinear mean value theorem of Pan–Ding–Wang [13, Lemma 2] and Motohashi [11, Theorem 1] for general convolutions with moduli up to $N^{1/2}(\log N)^{-B}$; see also [2, Theorem 0] for an explicit statement. $\square$

3. THE WEIGHTED VECTOR SIEVE

Fix parameters $0 < \alpha < 2\delta < 1/2$, $\delta < 1/4$, $0 < \lambda$, $\alpha < \beta_1 \leq 1/2$, and put

$$y = N^{1/2-\delta}, \quad z_0 = N^\alpha, \quad y_1 = N^{\beta_1}, \quad P = P(\mathcal{P}_3; y) P(\mathcal{P}_1; z_0).$$

Note that every odd prime $q < z_0$ divides P . For $n \in \mathcal{A}$ with $(n, P) = 1$ define

$$w_n = 1 - \lambda \sum_{\substack{q|n, q \in \mathcal{P}_1 \\ z_0 \leq q < y_1}} \left(1 - \frac{\log q}{\log y_1}\right), \quad W = \sum_{\substack{n \in \mathcal{A} \\ (n, P)=1}} w_n.$$

Lemma 3.1 (Structure of the survivors). *Let $n \in \mathcal{A}$ with $(n, P) = 1$. Then exactly one of the following holds:*

- (i) *all prime factors of n lie in $\mathcal{P}_1$, or $n = q^2 m$ with $q \in \mathcal{P}_3$; in both cases n is a sum of two squares;*
- (ii) *$n = q_1 q_2 m$ with $q_1 < q_2$, $q_1, q_2 \in \mathcal{P}_3$, $q_1 \geq y$, and $m \leq N^{2\delta}$ composed of primes of $\mathcal{P}_1$ that are $\geq z_0$.*

Proof. n is odd and $n \equiv 1 \pmod{4}$, so the number of prime factors of n in $\mathcal{P}_3$, counted with multiplicity, is even. All of them are $\geq y$, and $y^4 > N$ because $\delta < 1/4$; so there are 0 or 2 of them. If there are none, or the two coincide, n is a sum of two squares by Fermat's theorem. Otherwise we are in case (ii), and $m = n/(q_1 q_2) \leq N/y^2 = N^{2\delta}$; its prime factors are in $\mathcal{P}_1$ and $\geq z_0$ because $(n, P) = 1$. $\square$

Lemma 3.2 (Counting prime factors). *Let n be as in Lemma 3.1(i), squarefree, and suppose $w_n > 0$. Then $\Omega(n) < 1/\lambda + 1/\beta_1$.*

Proof. All prime factors of n are in $\mathcal{P}_1$ and are $\geq z_0$. Terms with $q \geq y_1$ satisfy $1 - \log q / \log y_1 \leq 0$, hence

$$\sum_{\substack{q|n \\ z_0 \leq q < y_1}} \left(1 - \frac{\log q}{\log y_1}\right) \geq \sum_{q|n} \left(1 - \frac{\log q}{\log y_1}\right) = \Omega(n) - \frac{\log n}{\log y_1} \geq \Omega(n) - \frac{1}{\beta_1}.$$

Thus $w_n > 0$ forces $\lambda(\Omega(n) - 1/\beta_1) < 1$. $\square$

The elements $n \in \mathcal{A}$ divisible by q^2 for some prime $q \geq z_0$ number at most $\sum_{z_0 \leq q \leq \sqrt{N}} (N/q^2 + 1) \ll N^{1-\alpha}$, which is negligible.

Proposition 3.3. *Let $\theta_3 \in (0, 1/2)$ and, for each $u \in [\alpha, \beta_1]$, let $t(u) \in (0, 1/2 - u)$. Put $s_3 = \theta_3/(1/2 - \delta)$, $s_1 = (1/2 - \theta_3)/\alpha$,*

$$\begin{aligned} L &= f(s_3)F(s_1) + F(s_3)f(s_1) - F(s_3)F(s_1), \\ R &= \frac{1}{2} \int_{\alpha}^{\beta_1} \left(1 - \frac{u}{\beta_1}\right) F\left(\frac{t(u)}{1/2 - \delta}\right) F\left(\frac{1/2 - u - t(u)}{\alpha}\right) \frac{du}{u}. \end{aligned}$$

Then

$$W \geq (L - \lambda R + o(1)) \frac{1}{2e^{\gamma} \sqrt{\alpha(1/2 - \delta)}} \mathfrak{S}(N) \frac{N}{(\log N)^2}.$$

Proof. Write $a = 1_{(n, P(\mathcal{P}_3; y))=1}$, $b = 1_{(n, P(\mathcal{P}_1; z_0))=1}$, and let $a^{\pm}, b^{\pm}$ be the sums of the semi-linear sieve weights of Lemma 2.2 of levels $D_3 = N^{\theta_3}$ and $D_1 = N^{1/2-\theta_3}(\log N)^{-B}$. By Lemma 2.3,

$$\sum_{\substack{n \in \mathcal{A} \\ (n, P)=1}} 1 \geq \sum_{n \in \mathcal{A}} (a^{-}b^{+} + a^{+}b^{-} - a^{+}b^{+}).$$

Since the prime sets are disjoint, $g(de) = g(d)g(e)$ for $d \mid P(\mathcal{P}_3; y)$, $e \mid P(\mathcal{P}_1; z_0)$, and

$$\sum_{n \in \mathcal{A}} a^{-}b^{+} = X \left(\sum_d \lambda_d^{-} g(d) \right) \left(\sum_e \lambda_e^{+} g(e) \right) + O \left(\sum_{m \leq D} \tau(m) |r_m| \right),$$

and similarly for the other two terms. By Lemma 2.2 the main terms combine to

$$XV(\mathcal{P}_3; y)V(\mathcal{P}_1; z_0)(L + o(1)),$$

and by Lemma 2.1 the error is $O(N(\log N)^{-A})$.

For the weights, fix $q \in \mathcal{P}_1$, $q = N^u$ with $\alpha \leq u < \beta_1$. By Lemma 2.3, the number of $n \in \mathcal{A}$ with $q \mid n$ and $(n, P) = 1$ is at most $\sum_{n \in \mathcal{A}, q|n} a_q^{+} b_q^{+}$, where a_q^{+}, b_q^{+} are upper weights of levels $N^{t(u)}$ and $N^{1/2-u-t(u)}(\log N)^{-B}$. This is

$$Xg(q)V(\mathcal{P}_3; y)V(\mathcal{P}_1; z_0) \left(F\left(\frac{t(u)}{1/2 - \delta}\right) F\left(\frac{1/2 - u - t(u)}{\alpha}\right) + o(1) \right) + O \left(\sum_{m \leq D/q} \tau(m) |r_{qm}| \right).$$

Summing over q , the error terms total $\ll \sum_{m \leq D} \tau_3(m) |r_m| \ll N(\log N)^{-A}$ by Lemma 2.1, and the prime number theorem modulo 4 turns $\sum_{q \in \mathcal{P}_1} g(q)(1 - \log q / \log y_1)(\cdots)$ into $R + o(1)$ (the

class $\mathcal{P}_1$ has density $1/2$). Since the integrand involves $F(s) \leq A/\sqrt{s}$ only as $s \rightarrow 0^+$, the integral converges.

Finally, by Mertens' theorem and its analogue in progressions modulo 4,

$$V(\mathcal{P}_3; y)V(\mathcal{P}_1; z_0) = \prod_{2 < q < z_0} (1 - g(q)) \prod_{\substack{q \in \mathcal{P}_3 \\ z_0 \leq q < y}} (1 - g(q)) \sim \frac{\mathfrak{S}(N)e^{-\gamma}}{\alpha \log N} \left(\frac{\alpha}{1/2 - \delta} \right)^{1/2},$$

and $X \sim N/(2 \log N)$, which gives the stated normalisation. $\square$

4. SWITCHING

Let $\mathcal{E}$ be the set of $p \leq N$, $p \equiv N - 1 \pmod{4}$, such that $n = N - p$ satisfies $(n, P) = 1$ and falls under Lemma 3.1(ii). Let $\mathcal{T}$ be the set of triples (q_1, q_2, m) with $q_1 < q_2$, $q_i \in \mathcal{P}_3$, $q_1 \geq y$, $q_1 q_2 m \leq N$, and m composed of primes of $\mathcal{P}_1$ that are $\geq z_0$ (so $m \leq N^{2\delta}$), and let $\mathcal{B}$ be the multiset $\{N - q_1 q_2 m : (q_1, q_2, m) \in \mathcal{T}\}$.

Proposition 4.1. *We have $\#\mathcal{E} \leq (4c(\delta, \alpha) + o(1))\mathfrak{S}(N)N/(\log N)^2$, where*

$$c(\delta, \alpha) = \frac{1}{4} \int_{[0, 2\delta]} \frac{1}{1 - \mu} \log \frac{1/2 + \delta - \mu}{1/2 - \delta} d\nu(\mu), \quad \nu = \sum_{j \geq 0} \frac{\nu_1^{*j}}{j!}, \quad d\nu_1(\mu) = \frac{1}{2} 1_{[\alpha, 2\delta]}(\mu) \frac{d\mu}{\mu},$$

(ν_1^{*0} is the unit mass at 0, and ν is restricted to $[0, 2\delta]$).

Proof. Every $p \in \mathcal{E}$ is a prime element of $\mathcal{B}$, and an element $b \in \mathcal{B}$ with $b \geq D = N^{1/2}(\log N)^{-B}$ that is prime is coprime to $P(D) = \prod_{2 < \ell < D} \ell$ (all elements of $\mathcal{B}$ are odd). Hence

$$\#\mathcal{E} \leq S(\mathcal{B}, D) + \#\{t \in \mathcal{T} : N - t < D\}, \quad S(\mathcal{B}, D) = \#\{b \in \mathcal{B} : (b, P(D)) = 1\}.$$

The last term is $\ll D N^\varepsilon$. For odd squarefree d write $\#\{b \in \mathcal{B} : d \mid b\} = |\mathcal{B}|g_{\mathcal{B}}(d) + r_{\mathcal{B}}(d)$ with $g_{\mathcal{B}}(\ell) = 1/(\ell - 1)$ for $\ell \nmid N$, $g_{\mathcal{B}}(\ell) = 0$ for $\ell \mid N$, $\ell < z_0$. (Primes $\ell \mid N$ with $\ell \geq z_0$ are at most $1/\alpha$ in number and change the main term by a factor $1 + o(1)$.) The sequence $t = q_1 \cdot (q_2 m)$ is a convolution with q_1 in $[N^{1/2-\delta}, N^{1/2}]$. We split the ranges of q_1, q_2 and m dyadically into $O((\log N)^3)$ boxes. In each box the conditions $q_1 < q_2$ and $q_1 q_2 m \leq N$ are removed by Perron's formula (or a smooth partition of unity), at the cost of a factor $(\log N)^C$ and an admissible error term; the resulting sequences are convolutions to which Lemma 2.4 applies, with q_1 carrying the Siegel–Walfisz condition. Taking A large in Lemma 2.4 we obtain $\sum_{d \leq D} |r_{\mathcal{B}}(d)| \ll N(\log N)^{-A}$. The upper bound linear sieve at level D with sifting range $z = D$, i.e. $s = 1$, gives

$$S(\mathcal{B}, D) \leq |\mathcal{B}| V_{\mathcal{B}}(D) (F_1(1) + o(1)) + O\left(\sum_{d \leq D} |r_{\mathcal{B}}(d)|\right), \quad F_1(1) = 2e^\gamma.$$

Here

$$V_{\mathcal{B}}(D) = \prod_{2 < \ell < D} (1 - g_{\mathcal{B}}(\ell)) \sim \frac{\mathfrak{S}(N)e^{-\gamma}}{\log D},$$

and $\log D \sim \frac{1}{2} \log N$, so $S(\mathcal{B}, D) \leq (4 + o(1))\mathfrak{S}(N)|\mathcal{B}|/\log N$.

It remains to evaluate $|\mathcal{B}| = |\mathcal{T}|$. Writing $m = N^\mu$, $q_1 = N^u$ and using the prime number theorem in progressions modulo 4 for q_1 and q_2 ,

$$|\mathcal{T}| = \sum_m \sum_{\substack{q_1 \in \mathcal{P}_3 \\ y \leq q_1 < \sqrt{N/m}}} \pi\left(\frac{N}{mq_1}; 4, 3\right) (1 + o(1)) = \frac{N}{\log N} \left(\sum_m \frac{1}{m} \cdot \frac{1}{4} \int_{1/2-\delta}^{(1-\mu)/2} \frac{du}{u(1-\mu-u)} + o(1) \right),$$

and $\int_{1/2-\delta}^{(1-\mu)/2} \frac{du}{u(1-\mu-u)} = \frac{1}{1-\mu} \log \frac{1/2+\delta-\mu}{1/2-\delta}$. The sum over m (products of j primes of $\mathcal{P}_1$ in $[z_0, N^{2\delta}]$, $j \geq 0$) is bounded above by the integral against ν (dropping the distinctness of the

prime factors only increases it), by Mertens' theorem modulo 4. This gives $|\mathcal{B}| \leq (c(\delta, \alpha) + o(1))N/\log N$. $\square$

5. NUMERICAL VERIFICATION AND PROOF OF THEOREM 1.1

By Lemma 3.1, the survivors split into the set of case (i), on which w_n is the weight we want, and the set of case (ii), which corresponds to $\mathcal{E}$. Since $w_n \leq 1$,

$$\sum_{\substack{n \in \mathcal{A}, (n, P)=1 \\ \text{case (i)}}} w_n \geq W - \#\mathcal{E}.$$

By Propositions 3.3 and 4.1 the right-hand side is at least $(\text{net} + o(1))\mathfrak{S}(N)N/(\log N)^2$, where

$$\text{net} = \frac{L - \lambda R}{2e^\gamma \sqrt{\alpha(1/2 - \delta)}} - 4c(\delta, \alpha).$$

We choose

$$\delta = \frac{1}{10}, \quad \alpha = \frac{7}{1000}, \quad \lambda = \frac{3}{20}, \quad \beta_1 = \frac{3}{10}, \quad \theta_3 = 0.478,$$

so that $1/\lambda + 1/\beta_1 = 20/3 + 10/3 = 10$ and, by Lemma 3.2, every squarefree case-(i) survivor with $w_n > 0$ has $\Omega(n) \leq 9$.

Lemma 5.1. *For these parameters, $L \geq 0.583795$, $R \leq 2.050122$, $c(\delta, \alpha) \leq 0.339858$ and $\text{net} \geq 0.106297$.*

Method of verification. The computation is carried out by the accompanying script `certify.py`.

- (1) On $(0, 2]$ and $[1, 3]$, F and f are evaluated from their closed forms in interval arithmetic (`mpmath.iv`, 80 bits). For $s \in [2, 80]$ the two delay equations are integrated on the grid $s \in 10^{-3}\mathbb{Z}$: on each step the lagged function is enclosed between its values at the two end points (by monotonicity), and $\int ds/(2\sqrt{s})$ is computed exactly. Each floating point operation is followed by one outward rounding step, which is sufficient since $+$, $-$, $\times$, $/$, $\sqrt{}$ are correctly rounded in IEEE 754. For non-grid arguments, monotonicity gives one-sided bounds from neighbouring grid points.
- (2) L is bounded below using $L = f(s_3)F(s_1) + F(s_3)(f(s_1) - F(s_1))$ and $f(s_1) - F(s_1) \leq 0$.
- (3) R is bounded above by a Riemann sum over 6000 cells, with $t(u)$ constant on each cell; on a cell $[u_a, u_b]$ the factor $(1 - u/\beta_1)/u$ is largest at u_a and $F((1/2 - u - t)/\alpha)$ is largest at u_b .
- (4) For c , the measure ν_1 is replaced by masses $\frac{1}{2} \log \frac{k+1}{k}$ placed at the left end points $k/2000$ of the cells; the convolution powers then dominate ν on initial segments. The integrand $I(\mu) = \frac{1}{1-\mu} \log \frac{0.6-\mu}{0.4}$ is decreasing on $[0, 0.2]$ (its derivative has the sign of $I(\mu) - 1/(0.6 - \mu) < 0$), so moving mass to the left gives an upper bound. The floating point convolution is multiplied by $1 + 10^{-9}$.

$\square$

Proof of Theorem 1.1. By Lemma 5.1 and the discussion above, for N large, $\sum_{\text{case (i)}} w_n \geq 0.106 \mathfrak{S}(N)N/(\log N)^2$. Removing the $\ll N^{1-\alpha}$ non-squarefree elements, each remaining n with $w_n > 0$ is a sum of two squares with $\Omega(n) \leq 9$, and since $w_n \leq 1$ the number of such n , i.e. of primes p , is at least $\frac{1}{10} \mathfrak{S}(N)N/(\log N)^2$. The constant N_0 is ineffective because of the Siegel–Walfisz theorem underlying Lemmas 2.1 and 2.4. $\square$

Remark 5.2. Floating point computations (not certified) indicate that without Richert's weights ($\lambda = 0$) the same method gives $\Omega(n) \leq 24$, and that with the switching constant 5 in place of 4 it still gives $\Omega(n) \leq 11$; see Figure 2.

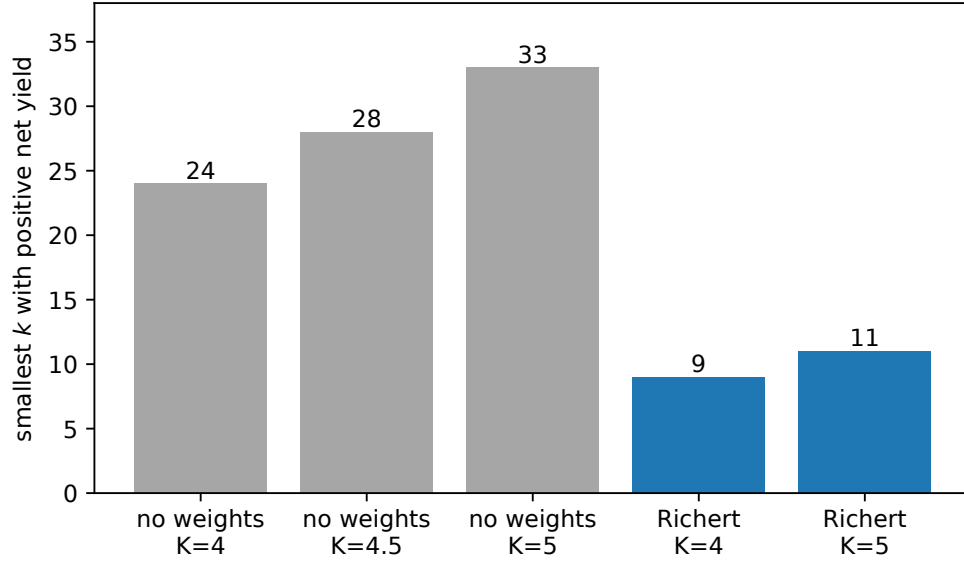


FIGURE 2. Smallest bound k for $\Omega(N - p)$ obtained by floating-point parameter scans, without weights and with Richert's weights, for several values of the switching constant K . Only the value $k = 9$ ($K = 4$, Richert's weights) is certified (Lemma 5.1); the other values are illustrative.

DATA AND CODE AVAILABILITY

The verification script `certify.py`, the exploratory scripts and their outputs are archived at [doi:10.5281/zenodo.23133054](https://doi.org/10.5281/zenodo.23133054).

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